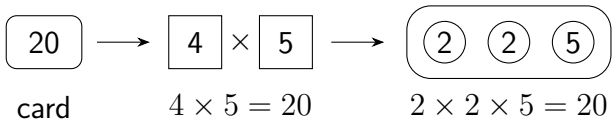
A number card can split into two whole numbers greater than 1 that multiply to its value. These numbers are its factors. Keep splitting until every card is prime: a number greater than 1 with no factors except 1 and itself. The finished cards make a prime bag. Multiply its tokens to find its value.



Token order does not matter. Equal tokens are interchangeable.

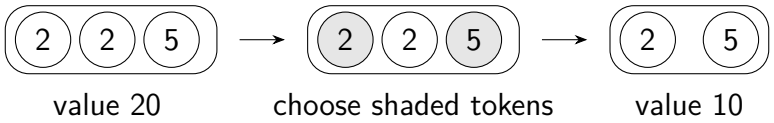
Problem 1: Find different splitting routes for 36, 48 and 60. Can a number have two different finished prime bags?

36

48

60

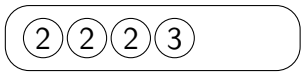
Choose some tokens from a bag, using each token at most once. The chosen tokens form a bag. You may choose all of them or none. An empty bag has value 1. A divisor of a whole number divides it with no remainder.



value 20 choose shaded tokens value 10

$2 \times 5 = 10$ $20 \div 10 = 2$

Problem 2: For each bag, find every different value you can make by choosing tokens. Explain why your collection is exactly the set of divisors of the full bag's value.



value 24



value 30

Problem 3: How many different values can you make by choosing tokens from each bag? Find a way to be certain of each total without listing every value.

(2)(2)(3)(3) value 36

(2)(2)(3)(5) value 60

(2)(2)(2)(3)(3) value 72

Problem 4: A bag has seven 2 tokens and four 3 tokens. How many different divisors does its value have? Explain a counting rule that works for any prime bag.

Problem 5: Draw prime bags using any of 2, 3 and 5 whose values have exactly 6, 8, 9, 10 and 12 divisors. Find all possible token-count patterns for each target. Bags with two 2s and one 3, or two 5s and one 2, have the same pattern.

6 divisors

8 divisors

9 divisors

10 divisors

12 divisors

Problem 6: Find the least positive number with each divisor count below. You may use any primes. Explain how you can rule out every smaller number.

4 divisors

6 divisors

8 divisors

9 divisors

10 divisors

12 divisors